Cracks in the knowledge: sea ice terms in Kangiqsualujjuaq, Nunavik

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In the Canadian Arctic, where sea ice is a feature of purposeful and spiritual significance to the Inuit inhabitants, evidence suggests that traditional knowledge of the terms for sea ice is diminishing. This article presents the findings from fieldwork regarding the terms, and their definitions, that a group of Inuit from Kangiqsualujjuaq in Nunavik, Northern Quebec, use to describe sea ice. This case study, which explores their knowledge of sea ice phenomena, offers insights into the present state of Inuit knowledge of sea ice terminology in that community. Instruments used to elicit knowledge of and about sea ice terms are described in this article. Participant inclusion was based on gender and age groups, and an illustration was produced to demonstrate the disparity in sea ice knowledge across three generations of male and female hunters. It is suggested that future nomenclature studies of Inuit knowledge be based on the canvassing of all age brackets, if a realistic picture of the ways in which such knowledge is transmitted within and across generations is to be ascertained.

Keywords: sea ice; Inuit knowledge, knowledge transmission, generations, Kangiqsualujjuaq, Kangiqsualujjuamiut, Inuktitut

Fissure dans les connaissances : les mots pour désigner la glace de mer à Kangiqsualujjuaq, Nunavik

Pour les Inuits, la glace de mer de l'océan Arctique canadien forme un élément particulièrement important d'un point de vue pragmatique et spirituel. Il ressort des données disponibles que les connaissances traditionnelles des mots utilisés pour décrire la glace de mer se perdent. Cet article présente les conclusions principales d'une étude menée sur le terrain et qui portait sur ces mots et les définitions données par un groupe d'Inuits vivant à Kangiqsualujjuaq, Nunavik dans le Nord du Québec pour décrire la glace de mer. L'objectif de cette étude de cas est d'explorer le champ des savoirs sur la glace de mer et d'expliciter l'étendue des connaissances actuelles des Inuits de la terminologie propre à la glace de mer. L'article décrit les instruments ayant servi à obtenir des renseignements sur la connaissance des mots pour désigner la glace de mer. Les deux critères d'inclusion retenus pour cette étude étaient le sexe et le groupe d'âge. Ainsi, le portrait dressé montre à quel point les champs des savoirs sont variables d'une génération à l'autre et selon le sexe des chasseurs. Nous en concluons que les études futures sur la nomenclature utilisée par les Inuits ne doivent exclure aucun groupe d'âge, afin d'obtenir une image complète des processus de transmission de ces connaissances au sein et entre les générations.

Mots clés : glace de mer, connaissances des Inuits, transfert des connaissances, générations, Kangiqsualujjuaq, Kangiqsualujjuamiut, Inuktitut

Introduction

On the eastern shores of Ungava Bay, the near-shore ice rises and falls in tune with extreme tides that reach up to 16 metres (Figure 1). Combined with perennial freeze-thaw action, this natural process makes the Ungava Coast one of the most dynamic and elastic settings in the world. The Inuit who live and hunt along these shores are the Kangiqsualujjuamiut (Inuit of Kangiqsualujjuaq). As littoral people, they have developed rich and varied ways of understanding this

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Figure 1
The Kangiqsualujjuaq coast at low tide in late winter 2007. The sea ice that forms within the estuary of the George River and along the eastern seaboard of Ungava Bay moves with the tides, and is constantly reconfigured through wave action. This region, like the Bay of Fundy, is subject to some of the most extreme tidal fluctuations in the world

coastal environment over the course of more than a thousand years.

The coast is central to Kangiqsualujjuamiut identity and sense of place. But as this article suggests, their connection to the coast may be changing, due to residents' decreasing knowledge of their coastal environment. Although this article focuses on sea ice, it is based on a wider study of Inuit conceptions of the coast that suggests other forms of coastal knowledge are also in decline (Heyes 2007). These include Inuit knowledge of coastal nomenclature, place names, spatial terminology, and maritime myths and legends (Heyes 2007; Heyes and Jacobs 2008). As discussed in subsequent sections, this is largely due to ongoing changes to the physical landscape and to the social milieu, along with changes in the ways that Inuit learn traditional skills. Drawing on fieldwork carried out from 2003 to 2008 on Kangiqsualujjuamiut uses of terms relating to sea ice, this article discusses why such terms are only partially transmitted to other community members and explores the impacts this may have on Inuit attachment to the coast. The discussion is based on information obtained from 34 Inuit men and women selected from six family units within the community of Kangiqsualujjuaq (Figure 2).

The first section of this article explores the practical, as well as the mythical and spiritual, qualities of sea ice, and a discussion of the factors that shape sea ice knowledge. The second section describes the methods and instruments of the study, while the third section comprises a discussion of the past, current, and possible future of sea ice knowledge and nomenclature within the local Inuit community.

Sea Ice as Place and Legend

Sea ice, or *tuvaq* to the Kangiqsualujjuamiut, is much more than simply a form of water. It is an enchanting and 'astonishing substance', as Gosnell (2005) describes in the title of her book on ice. To the Inuit, who know the sea ice and its intricacies—perhaps better than any other society—this substance possesses both functional and spiritual qualities (Heyes and Jacobs 2008). Although the Inuit appreciate the aesthetic qualities of sea ice, it is also: a medium for travel; a platform from which to hunt sea animals; a building material (Rapoport 1969; Oliver 2003); a navigational aid (Mathiassen 1928; Spink and Moodie 1972; Fortescue 1988; Aporta and Higgs 2005); an inspiration for artistic endeavours (Etok 1975, 1976); a place for learning and transmitting knowledge and a space that harbours creation stories and legends (Metayer 1972; Shaw 2002). Sea ice gives life and can take life. It sings and creaks, bends and warps in response to wind and tides. Depending on the way light casts upon it, sea ice can be transformed into shades of white, grey, blue, or turquoise. Poetics aside, there are few natural features imbued with as much meaning and significance to

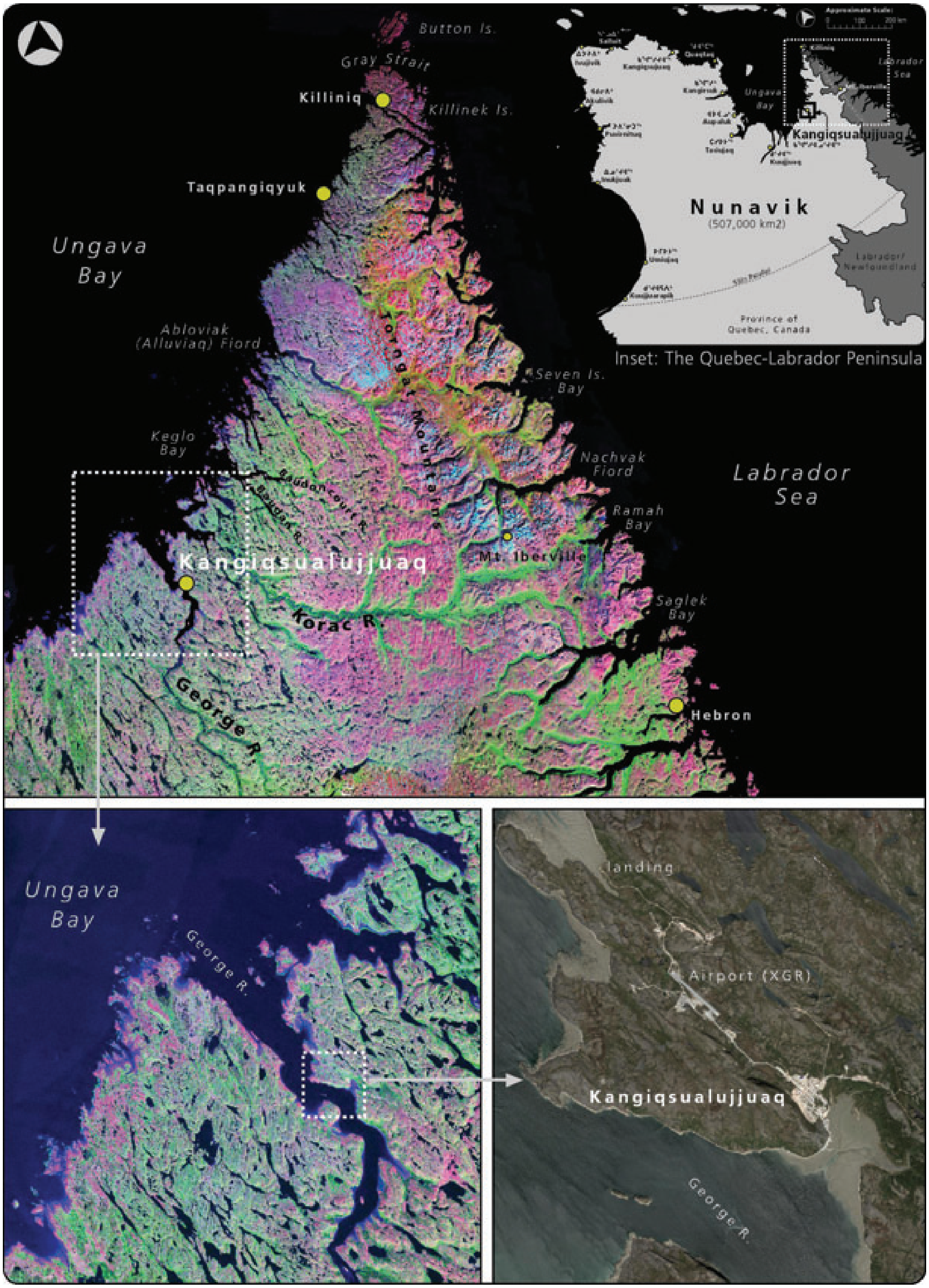

Figure 2
Location of the village of Kangiqsualujjuaq within the Quebec-Labrador Peninsula and against the George River. The village straddles the tree line and tundra and is only accessible from the south by ship or plane. The popular hunting locations of Killiniq, Abloviak Fiord and Nachvak Fiord lie north of Kangiqsualujjuaq

the Kangiqsualujjuamiut as this complex, unpredictable medium. It comes in many forms that have been named by the Kangiqsualujjuamiut, names that are used to distinguish and classify sea-space.

These terms should be discussed in context with how this substance is used by the Kangiqsualujjuamiut. There are several surfaces to sea ice, in horizontal as well as vertical dimensions, that are utilized by these people. For example, Kangiqsualujjuamiut rely upon caverns that form between the cracks in sea ice (*ilumaniq*). Although dangerous, hunters use these openings to access the seabed in order to collect clams and seaweed at low tide. Another sea ice surface, called *qarvitinniq*, generally forms along the shore or a cliff edge. It forms a platform of ice that the Kangiqsualujjuamiut can travel on by snowmobile in late spring. At this time of year, this ice affords safer travel than the ice that forms further seaward, as it is usually detached or raised from *tuvaq*, and does not contain as many hidden cracks and crevices.

While having myriad pragmatic uses, sea ice is also bound up with Kangiqsualujjuamiut cosmology. In a 1985 interview, Elder Noah Angnatuk recounted that the break-up of sea ice, and the formation of large waves along the coast, was caused by the mythical *Nanukulluk*, a giant polar bear, moving his body (Avataq Cultural Institute 1985). Elder Tivi Etok has made prints and carvings based on Kangiqsualujjuamiut stories (Etok 1975, 1976), many of which depict legendary beings roaming the sea ice. One such narrative is about the *Mitilik,* downcovered creatures confined to the sea ice domain. Etok said that the *Mitilik* could metamorphose into any animal (particularly seals), and would attack hunters walking the ice in search of seals basking beside their breathing holes (Heyes 2007).

This notion of sea ice as a dwelling place of nonhumans is also evident in the story of *Sikuliasiujuitu*, by Paul Jararuse of Kangiqsualujjuaq. Literally, 'unable to go on thin ice', *Sikuliasiujuitu* refers to a giant who was too heavy to travel on sea ice to hunt seals. Becoming hungry and frustrated, he would steal the catch of hunters returning to camp. *Sikuliasiujuitu's* unwillingness to share the catch led the hunters to take revenge in a brutal attack (Heyes 2007). Other beings, called *Inugagullit* (small people), are said to frequent the coastal region, including the sea ice. Less than half a metre tall, *Inugagullit* hunt sea mammals using kayaks, harpoons, and lances. Stories describe how they catch bearded seals and then prepare the seal meat at the floe edge. This zone of sea ice also appears to be connected to spirits. Tivi Etok recounted being followed by an evil dark cloud when hunting along the floe edge with his father. He told of how he was taunted by voices from the cloud, and that it tried to collect his soul (Heyes 2007). These stories told by Kangiqsualujjuamiut Elders indicate that the sea ice around Kangiqsualujjuaq continues to be considered a place where nonhuman entities exist.

Sea Ice as a Marker of Time

Aside from sea ice's mythological associations, its presence or absence is entwined with Inuit concepts of time and season (Freeman 1984; McDonald *et al*. 1997; Aporta 2002). The emergence of sea ice, like autumn's coloured leaves in warmer climates, is a seasonal indicator (MacDonald 2000). In Kangiqsualujjuaq, the formation of sea ice is greeted with joy, as it signals the arrival of winter and the activities associated with the winter months, such as recreational dog-team travel and ice-fishing. It is a time to mend nets, machines and sleds; a time that gives many hunters the opportunity to travel farther and wider on land than is possible during summer. Rivers that are otherwise inaccessible can be reached by travelling along shores and estuaries of the frozen sea. And, journeys to ancestral hunting grounds in distant places such as Killiniq, Abloviak Fiord, and Nachvak Fiord (see Figure 1) are embarked upon shortly after the sea ice is thick enough to support a snowmobile and sled.

As much as winter is favoured, the Kangiqsualujjuamiut welcome the onset of other seasons as well. The collapse of *akluk* (breathing holes) and the appearance of *uttuq* (basking seals), which indicate a shift to springtime, is cause for celebration among hunters. It heralds an end to the bitter cold and the return of geese and other waterfowl from the South. Sunlight increases and boats return to the water. With the break-up of river and near-shore ice, char and salmon begin their downstream runs. Many families camp along shores and inlets during the break-up of sea ice, in order to hunt seals and other animals. Summer is usually marked by the full break-up of sea ice. When it has gone and the animals have dispersed, the shore is largely vacated by the Inuit. The sea ice is replaced by *tariuk* (the sea), which supported it and lay hidden beneath it. Hunting activities in the near-shore environment will resume when the sea ice builds again, months later.

Contextualizing Sea Ice Knowledge

Living in this coastal setting for more than a millennium, the Kangiqsualujjuamiut have collected knowledge requiring nomenclature for the sea and its various manifestations. This knowledge encompasses: place names for features of the land and frozen sea (Müller-Wille 1987); rich mythologies and creation stories associated with the sea and sea ice (Kohlmeister and Kmoch 1814; Anonymous 1833; McLean 1849; Hawkes 1916; Hantzsch 1932; Turner 2001; Heyes and Jacobs 2008); the location and movement of sea animals and birds; weather patterns (Makivik Corporation 1985); and survival and wayfinding skills. A concept-mapping diagram (Figure 3) exemplifies the breadth and the connectedness of Inuit forms of knowledge. Such concept maps, designed to picture Inuit knowledge associated with sea ice, were generated in order to better understand the relationships of subjects, and the flow of information about them, that emerged during interviews with community members.

Prior to the 1950s, when the Kangiqsualujjuamiut still lived in separate hunting camps scattered along the coast[1], knowledge was transmitted orally. This was a time when Inuit customs and traditions were passed on predominantly amongst hunters, through hunting-based activities while on the land, and also around the igloo. The outdoors was the classroom and the more experienced hunters were the teachers. Knowledge of sea ice, like other forms of knowledge, was gained through personal experience and tutelage. Since Inuit men and women hunted and travelled on the ice from an early age, they were obliged to know the nuances and patterns of sea ice and its related phenomena in order to survive. They therefore needed to internalize and retain an intimate knowledge of the ice, and needed ways to pass this information on to others.

Young and middle-aged Inuit now have difficulty attaining the same knowledge of their surroundings in as great detail and have difficulty articulating this knowledge due to changes in the way that knowledge is being transmitted within their communities. The younger generations have been taught in classrooms, principally by non-Inuit educators. Although the children are now taught exclusively in Inuktitut until grade 3 in Nunavik, a practice that may somewhat mitigate the loss of localized knowledge, Inuit students of all ages spend the majority of their school hours indoors; field-based activities form only a portion of the curriculum. This has had a profound effect on the ability to memorize and recall information about the land and the sea. Many Kangiqsualujjuamiut maintain that such knowledge is retained better when taught outside, in a participatory environment. Indeed, those who regularly hunt with their families during evenings and weekends seem to have a better grasp of knowledge pertaining to the land and sea than do those who are unable (or have little desire) to venture out on hunts.

While Inuktitut might be regarded as a resilient language (because it is being upheld in Kangiqsualujjuaq as the mother tongue), the extent to which subtleties of the language are grasped and retained by the young is largely affected by their learning environments, social relations, and willingness to learn. These challenges, together with the wide use of English in the village, have caused their Inuktitut comprehension skills to suffer. Consequently, their knowledge of the stories and terms regarding sea ice has been adversely affected, along with other knowledge traditionally communicated in Inuktitut.

Another factor contributing to the decline in Inuktitut is negative peer pressure regarding scholastic excellence. Often, those with a keen interest in learning within the school system

[1] With the support of the Canadian Government, the Inuit from the Quebec-Labrador Peninsula started constructing the village of Kangiqsualujjuaq in 1962, and permanently settled there in 1964. The creation of this centralized community, while initiated by the Government, was supported by the local Inuit (Arbess 1966). The 'settlement period' refers to the time after the establishment of villages in Nunavik.

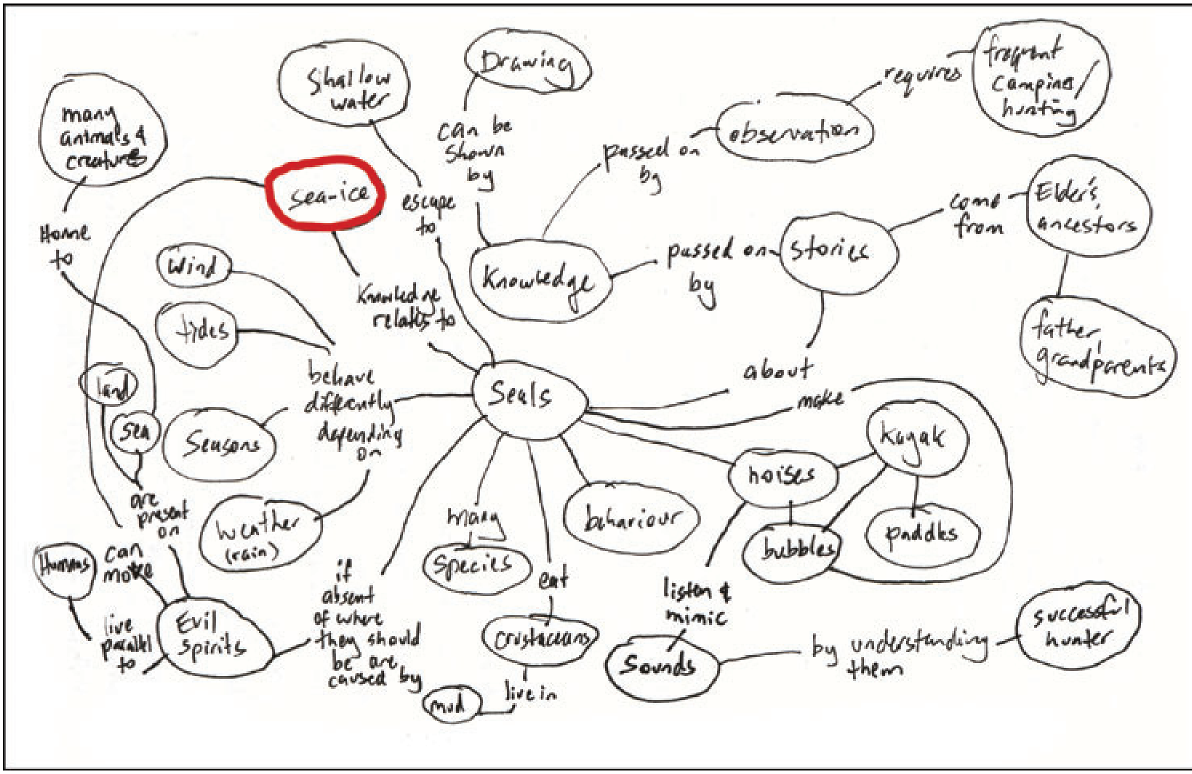

Figure 3

A concept map from 2005, showing how sea ice (highlighted) is situated within Kangiqsualujjuamiut knowledge frameworks. This was drawn by the author immediately upon the completion of an interview with an Elder. Following Novak and Gowin (1984), similar diagrams were made at the end of interviews with other hunters. Such concept maps indicate the significance of sea ice to hunters, and the ways in which the ice is connected to other features of the environment

are not as highly regarded among the youth as are, say, astute hunters of the same age. This discourages those who might want to pursue a formal education. And, although hunting provides one good avenue for learning Inuktitut, a scholastic path also helps the young gain a greater appreciation of the language. This is because most of the knowledgeable Inuktitut speakers in the community are also teachers at the school; by attending school, students have more opportunities to spend time with these Inuktitut experts than they would otherwise.

Other factors, such as a lack of engagement with Elders[2], and the high costs of hunting and travelling on land and sea[3], must also be kept in mind when considering the differences

[2]In Kangiqsualujjuaq, an individual may or may not be regarded as an Elder on the basis of age alone, though in most instances, an individual does become an Elder when he reaches 50–55 years. For Kangiqsualujjuamiut, Elder is a title given to individuals in recognition of the leadership and wisdom they provide to the community. While Elders are, by definition, custodians of and experts on Inuit knowledge, other younger individuals may also be considered very wise and knowledgeable and be held in high regard as outstanding members of the community.

[3]Many hunters' children are eager to participate in hunts, but are unable to do so because there are not sufficient modes of transport or supplies to accommodate them. While the Hunter-Support Programme in Kangiqsualujjuaq subsidizes some hunting equipment and provides loans to hunters, a growing population and limited funds means it remains difficult for young Inuit hunters to purchase tools of the trade such as snowmobiles, boats, and all-terrain vehicles.

in knowledge of sea ice terminology that exist between generations. As observed by the author, the social dynamics in the community are such that middle-aged hunters with the most knowledge of sea ice largely hunt and socialize together. Since it is difficult for new hunters to join well-established hunting partnerships, in many cases knowledge of sea ice, like other hunting-related knowledge, is confined to a circle of a few hunters. Partially offsetting this, attempts have been made within the community to encourage the dissemination of hunting knowledge from learned hunters through participatory learning. For instance, Elders frequently discuss hunting on the community radio station. Activities are also arranged by the local school, whereby the Elders communicate their knowledge of the land through hunting-based modules. Elders have also been instrumental in developing and supporting the 'Individualized Paths of Learning' programme, which is affiliated with the school. This programme encourages disenfranchised or scholastically challenged students to learn land-based skills, such as how to operate dog teams, build igloos, and make snowshoes and hunting tools. However, it should be noted that despite all this, few additional interactions occur between Inuit youth and Elders outside of such forums[4]. This lack of dialogue between Elders and Inuit youth is associated with the complex and changing nature of intergenerational dynamics in the community (e.g., the impacts of modernization and social fragmentation). It will ultimately affect the capacity for Inuit youth to learn and retain hunting knowledge about systems such as sea ice.

This study was designed with these knowledge-transmission factors and agencies in mind. The methodology that was developed provides a basis for substantiated comparisons about the levels of sea ice knowledge among different generations of Kangiqsualujjuamiut.

[4]Local teachers, Elders, and middle-aged hunters all expressed concern about the younger members of the community having limited understanding of hunting knowledge (including sea ice knowledge). This issue is also actively being addressed in forums outside the community through such activities as the Junior Rangers Programme, the Ivakkak dog sled competition and internships and summer camps sponsored by the Avataq Cultural Institute.

Methods and Instruments

The wider study of Inuit knowledge and perceptions concerning the land-water interface, from which the results in this article on sea ice knowledge are drawn, was based on an ethnographic approach (Heyes 2007). Participant observation formed a significant portion of the study, along with nomenclature, drawing and mapping exercises. Over a fortnight in winter 2003, an investigative study framed around semistructured interviews was carried out in Kangiqsualujjuaq on the topic of Inuit myths and legends. A total of 25 Elders, middle-aged hunters, and Inuit secondary school students participated in that study. With this introduction to the people, the community and some of their traditional stories, the author returned to Kangiqsualujjuaq over three months in the spring and winter of 2004, and again for the same period in 2005, in order to focus specifically on Inuit knowledge of the coastal zone. Additional fieldwork was carried out in the winters of 2007 and 2008. Although a variety of instruments were used in the studies from 2003 to 2008, only those applicable to the elicitation of sea ice terms and coastal knowledge are discussed below.

Selection of the cohort

The cohort was based on age brackets, as exemplified by Kan's (1989) study of Tlingit perceptions of the Potlatch, and Krupnik and Vakhtin's (1997) exploration of Yupik concepts of reincarnation and animal-human interactions with the supernatural world. Longfellow in 1978 used a generational approach to analyze the productivity of the American workforce based on the notion of enculturation (Longfellow 1978; Scollon and Scollon 1995). This work also helped formulate the generational framework. In light of these studies, the rationale for the age brackets was based on the criteria outlined in Table 1.

Since knowledge transmission was the core focus of the study, the cohort was based on the selection of Inuit families regarded by the community as knowing more hunting traditions and stories of the land and sea than other families. Only regularly active hunting families were invited to participate, because it was assumed that less-active hunting families would

Table 1
Definitions of the age brackets that formed the basis of the research cohort

Generation		Definition
A	Elders (>55 yrs)	Those who were young children when villages were under construction or who grew up in family camps before villages were established in the 1950s. They did not attend formal schools. In their youth, they relied primarily on subsistence hunting for survival.
B	Middle-aged (31–54 yrs)	The first Inuit generation to live in villages throughout life. This is the first generation to attend formal elementary school. Some of this generation also attended residential high schools.
C	Youth (<30 yrs)	Those with the opportunity to attend both elementary and secondary schools. Significantly, they have grown up in the modern/electronic age and are proficient in and (like other Canadian youth) fascinated by emerging technologies and media. Furthermore, they were raised when land-claim agreements divided their homeland into regions based on hunting regulations and priorities.[a]

[a]For example, the James Bay Northern Quebec Agreement of 1975 (Canada 2004), which was the most significant land-claim agreement regarding Nunavik, reshaped the physical and political boundaries of Nunavik. Among other things, the Agreement divided the Nunavik Inuit homeland into three land regimes based on Inuit and non-Inuit hunting exclusivity (these are known as Category Lands). Prior to this Agreement, the Nunavik homeland was not subject to partitioning in any formal or legal sense. Recently, the development of a number of mining operations and provincial park systems has impacted the ways that the Inuit use the land, and conceptualize it. The impacts of this growth of the mining and protected-area sectors on Nunavik Inuit are yet to be fully understood.

not provide sufficient information about local environmental knowledge to permit robust analysis and discussion of how the environment is conceptualized within and among families. Accordingly, six hunting families were selected, each originating from a different region of the Quebec-Labrador Peninsula prior to the formation of the village in the 1960s[5]. This approach sought to account for inconsistencies in knowledge and perceptions of the coast that might be based on geographical differences in their places of origin. Moreover, the inclusion of several family units was designed to account for forms of knowledge, and ways of seeing the coast, that might be restricted to specific hunting families.

In light of the variability of family structures in Kangiqsualujjuaq, as well as the variability of hunting knowledge between family members, this project sought to solicit (where possible) the eldest male and female from each family unit, their eldest son and daughter and their eldest male and female grandchildren. In cases where a male or female member did not exist (or was deceased), the oldest two members of each generation were sought. Based on these criteria, 34 individuals, representing nearly five percent of Kangiqsualujjuaq's population, agreed to participate in the study. This cohort was composed of nine Elders (average age 72 years; 17 percent of the population of Elders), 12 middle-aged hunters (average age 43 years; 6 percent of the population of middle-aged hunters) and 13 Inuit youth (average age 22 years; 3 percent of population of their generation). This youngest generation comprises the largest proportion of the population of Kangiqsualujjuaq (64.3 percent as of 2005). An intergenerational research cohort, with equal gender representation, was considered to offer a more complete view of the range of knowledge and perceptions among the Kangiqsualujjuamiut, compared to interviewing only one group or one gender of knowledgeable hunters.

Semistructured interviews and knowledge trees

The 34 male and female hunters were interviewed, with the assistance of local translators, on several occasions, about their knowledge of the coast. Discussion topics ranged from notions and boundaries of place, to traditional hunting knowledge of the land and the sea, travelling and navigational practices, spatial concepts, nomenclature of coastal phenomena and traditional Inuit narratives. With a set of 30 questions relating specifically to the land-water interface, the interviews were also designed to discover whether such knowledge was being lost, gained, transformed, or enriched, in light of pedagogical

[5]Ancestral places of origin of the cohort include camps and villages located from Northern Labrador (Nain, Hebron, Makkovik, Nachvak Fiord), to the eastern shores of Ungava Bay (Killiniq, Taqpangiqyuk, Korac River, Kangiqsualujjuaq), to southern Ungava Bay (Kuujjuaq, Tasiujaq).

changes occurring within and from outside the community. Satellite imagery and hydrographic and topographic maps for the entire Quebec-Labrador Peninsula were used in the interviews. Also included were genealogical charts for each family unit.[6] Participants indicated, by drawing lines and arrows on the charts, the hunters from whom they learned the majority of their traditional knowledge and the hunters to whom they transmitted their traditional knowledge. The resulting web of lines illustrated the flow of knowledge within and across family units. These webs, or knowledge trees, revealed which hunters had been given much traditional knowledge through tutelage, as well as those who were not integral parts of the web of knowledge. The knowledge trees also provided opportunities to explore the extent to which individuals acquired knowledge of the environment on their own. However, the consensus among participants indicated that historically, as seems to be the case now, hunters have learned more about the land and the sea from interacting with each other, than from their own experiences in the environment.

Drawing instruments

The second phase of this study, described above, took place in 2004. The third phase, carried out in 2005, involved the use of drawing instruments. These were developed in response to the author's need to better understand the spatial aspects and other qualities of the coast that had been described by participants during the 2004 field trip. Following studies on land units by Calamia (1999) and Johnson (2000), who demonstrated that some indigenous groups compartmentalize space through the use of phenomena, toponyms, and nomenclature, this study attempted to establish whether the Inuit have ways of spatially classifying the land-water interface.

Two templates, shown in Figure 4, were created to obtain the Inuktitut terms that participants used to describe various phenomena, features, and geographic units of the seashore environment. To get the exercise underway, they were asked to record their terms for: tides; ice-free water; water at the change of the tides; water when frozen; alongshore features; offshore features and seasonal and ephemeral phenomena that appear on the tidal flats and offshore zones. Once the participants became familiar with the purpose of the exercise, they began to describe, and draw in detail, other coastal features that lay beyond the bounds depicted on the sheets. The templates, drawn in perspective and cross-section, and resembling a profile of the coast near the community, proved to be very useful tools for ascertaining nomenclature of the coast.[7] The exercise successfully exposed the breadth, and the limitations, of knowledge of the land-water interface among families and individuals. It also revealed the conventions by which coastal phenomena are named within and across the three generations.

The exercise was based on the supposition that those who could describe more coastal features likely possessed greater knowledge of the land-water interface than those who identified fewer phenomena. The names of coastal phenomena provided by each participant were categorized according to whether the nomenclature referred to: tides; water; ice; snow; coastal forms; spatial characteristics; atmospheric conditions; vegetation; activities; animals or vertical elements of the coast. Comparison of the naming conventions within the cohort was facilitated by tabulating the nomenclature data in a database, which enabled the data to be sorted by participant age, gender, family unit or generation.

Most participants wrote the terms they understood or recognized directly onto the templates; some drew sketches on the templates to further explain particular nomenclature or phenomena. Where possible, the Inuktitut orthography was later translated into the English equivalent, and then tabulated so that the identified terms could

[6] These charts were generated by the author using information gathered from interviews as well as from genealogical databases for Nunavik accessed at the Avataq Cultural Institute.

[7] The author was aware that a predetermined template might influence the way in which participants approached the drawing exercise. However, a pilot study, using a blank piece of paper, proved less successful than the template. A local translator was present to explain the nature of the templates in Inuktitut. Owing to the visual nature of the templates, the exercise was generally well understood by all participants, and cultural difficulties comprehending the templates were not encountered. For further details of how the template was designed, as well as the way the content was analysed, refer to Heyes 2007.

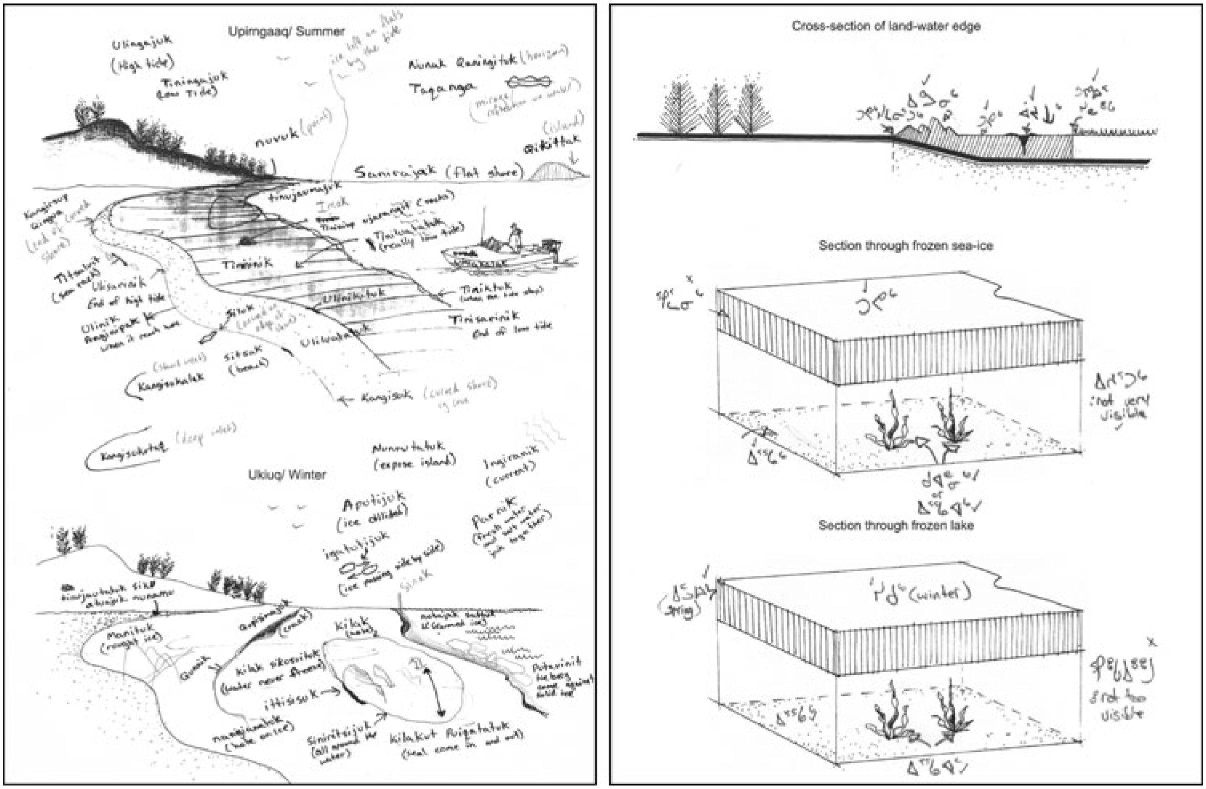

Figure 4
Inuktitut nomenclature of the coast, including sea ice. Pictured are templates (reduced from 11″ × 17″) generated for participants to identify, and discuss Inuktitut terms for, features and phenomena of the coastal environment. The templates, such as the sample pictured here, were drawn on by participants during face-to-face interviews. The interviews were carried out by the author, with the assistance of Inuktitut-speaking translators from Kangiqsualujjuaq

be compared according to participant generation and gender. (Local Inuktitut speakers assisted with the translation and transcription of these terms. Because many Inuktitut words are phonetically similar, the documenting of the interview sessions on digital recorders enabled transcribers to confirm the spelling of certain nomenclature from pronunciation).

Local Inuktitut translators provided orthography and definitions of sea ice terms used during the interviews if the participants did not commit them to paper themselves. Recognizing that translators might adjust terms conveyed by the participants to conform to their own understandings, translators were carefully briefed to record the terms verbatim. To avoid linguistic or cognitive misunderstandings between translators, participants, and the author, the visual aids described above were used to elicit knowledge wherever possible. This proved to be particularly effective, as locations and descriptions of sea ice features could be identified, or drawn directly on the visual aids. However, it should be noted that the descriptions in this article are the author's own interpretations of the information conveyed by the participants. These interpretations are based on oral histories and other relevant literature, information about sea ice conveyed directly by English-speaking Kangiqsualujjuamiut and indirectly through translators and interactions with hunters in the sea ice environment.

Figure 5
Inuit sea ice experts describe their knowledge of Inuktitut sea ice terms. Interviews were carried out in the field while overlooking the coast in early winter, and indoors while looking at slideshows of sea ice pictures. With the author are sea ice experts Daniel Annanack, Maggie Lucy Annanack, and Elder Tivi Etok

Photographic instruments

Although the nomenclature data were largely gathered during the 2004–2005 period, a series of follow-up photographic exercises were carried out with some Elders and middle-aged hunters from the original cohort, in 2007 and 2008. (It was not feasible to conduct the exercises with the entire cohort, due to budget and time constraints.) The selected individuals had been identified as having richer knowledge of the sea ice environment than the others. These 'sea ice experts', three of whom are pictured in Figure 5, were shown a slideshow of over 130 pictures of sea ice taken from alongshore and offshore regions of Kangiqsualujjuaq. Many pictures of the same ice forms were included in the set, in order to triangulate for consistency in the responses. The various forms of sea ice identified were recorded and discussed. Drawings and gestures were used extensively by the experts to describe the types and locations of ice formations. As in the earlier study, the terminology and physical properties of several ice features were elicited and then entered into the nomenclature database.

Making the most of winter's onset in December 2008, a number of the experts were brought to a landing beside the coast, where ice activity was visible. They were asked to identify the passing ice features. (The limitation of this approach in eliciting most or all of the sea ice terms known to the Kangiqsualujjuamiut, is that some forms of ice might not be present until later in the season. However, comprehensive discussions concerning the sea ice cycle from formation to breakup, suggested that a full picture of sea ice terminology was obtained while in the field).

The semistructured interviews, drawing exercises, knowledge trees, field exercises, and photographic exercises provided a range of ways to infer how fragile sea ice knowledge might be within the community. This multiplicity of evidentiary types, combined with the intergenerational approach, and the resulting information about sea ice terminology, provided a strong foundation for discussions of Inuit sea ice knowledge, and the mechanisms by which the people transmit this knowledge[8].

Results and Discussion

The nomenclature study from which this article on sea ice is drawn reveals that the Kangiqsualujjuamiut have at least 152 English-equivalent terms for describing facets of the coastal environment. Although sea ice terms featured prominently in this lexicon, numerous terms relating to tidal phenomena, geographic phenomena, water, ice, coastal vegetation, and spatial regimes were also represented. Of this coastal lexicon, it appears that only 51 of the 152 equivalent terms were previously recorded. Apparently, the other 101 terms relating to the coast identified by the cohort have not been documented (see Heyes 2007, 233). These previously unrecorded terms are significant, in that they might indicate that Kangiqsualujjuamiut terminology has expanded (e.g., through language hybridization or by the development of new terms) since Schneider (1985) generated a dictionary for the region. Or perhaps such terms were in use, but were not recorded when Schneider's dictionary was compiled, because of the project's limited scope at the time. In any case, these new terms provide opportunities to explore Kangiqsualujjuamiut knowledge systems of the coastal environment in more comprehensive ways than previously possible.

[8]The findings and the methodological approach of this study might apply within other contexts, such as in linguistic studies of Inuktitut. They might also provide insight for studies into Inuit navigation and wayfinding, as well as other Inuit knowledge systems associated with sea ice, such as climate variability, notions of season and the patterns, habits, and migration systems of sea birds and sea mammals.

Illustrating sea ice terms

From the 152 terms identified in the coastal lexicon study, 71 refer to sea ice forms or phenomena. A selection of these terms is illustrated in Figure 6, which shows the detail with which the Kangiqsualujjuamiut appreciate the rich and varied topography of sea ice. Figure 6 also shows the spatial dimensions of sea ice and its various manifestations with a sense of scale and form. Photographs, as well as sketches that the participants drew on the templates illustrated in Figure 4, were used as a guide to generate the illustrations. The features illustrated in Figure 6 are based on drawings of the coast that participants generated for other components of the study. Figure 6 encapsulates sea ice features that reach from the shoreline to the floe edge, a distance of 20 to 30 kilometres or more. To represent as many sea ice features as possible, it depicts a scene in late winter/early spring when a variety of sea ice features are present and the Inuit are actively engaging with this environment.

The orthography and definitions of sea ice terms that appear in Table 2 support Figure 6. The variations in terminology and usage were retained to reveal any individual or generational disparities regarding the uses and definitions of terms for the same ice feature. Where applicable, the orthography of sea ice terms that had been recorded in the most regionally relevant Inuktitut dictionary, *Ulirnaisigutit* (Schneider 1985), appear in Tables 2 and 3 for comparative purposes. The terms provided by the participants were not adjusted to conform to the orthography and definitions of *Ulirnaisigutit* for several reasons: first, Inuktitut spelling conventions are far from standardized; the syllabary lacks the sophistication necessary to capture all the sounds and subtleties of Inuktitut, which reflects the fact that Inuktitut remains primarily a spoken language; second, the experts consulted in the making of the *Ulirnaisigutit* dictionary were from different landscape and seascape settings to that experienced in Kangiqsualujjuaq, which may suggest a regional bias of the recorded terms, and third, *Ulirnaisigutit* was broad in scope—when it was devised, it did not focus on identifying terms

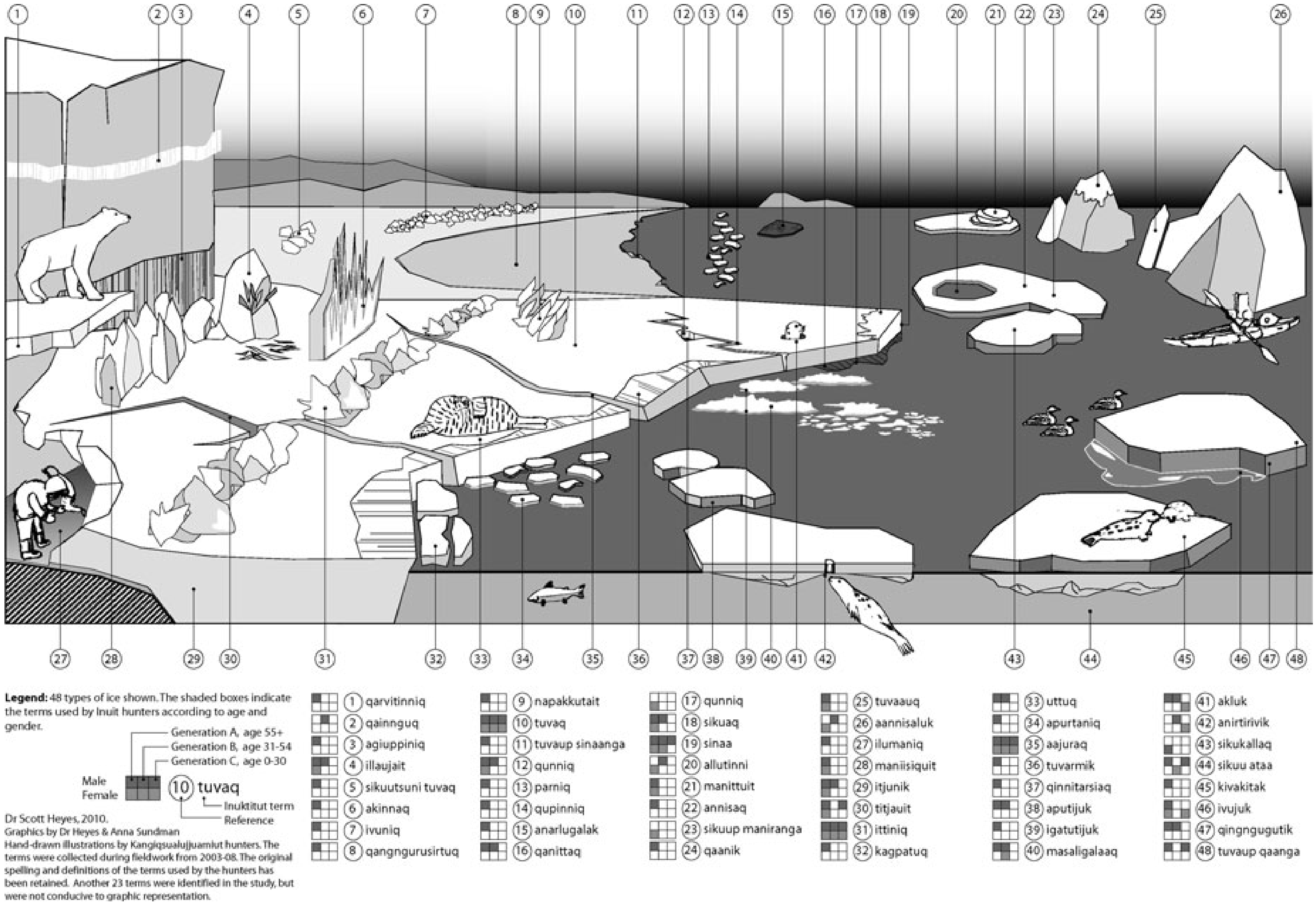

Figure 6
Depiction of Inuktitut terms (or their alternatives) as used by three generations of Kangiqsualujjuamiut hunters to describe late winter/early spring sea ice

Table 2
Definitions of sea ice terms depicted in Figure 6. Terms 49–71, marked with asterisks (*), were not conducive to graphic representation in Figure 6

Generation				Abbreviations		Note on Orthography
1	A	Male	>55 yrs	*Var*	Variant, where applicable	The term in bold typeface is either the term identified by more than one participant or the term provided by the most senior generation (where multiple terms and variations are provided by more than one generation). The original spelling of the terms used by the hunters has been retained. Such spellings and definitions may not be consistent with those in Inuktitut dictionaries and those in common usage in communities other than Kangiqsualujjuaq.
2	A	Female	>55 yrs	*Alt*	Alternative term, where applicable	
3	B	Male	31–54 yrs			
4	B	Female	31–54 yrs	(Sch)	Equivalent term(s) for Kangiqsualujjuaq region from Schneider (1985)	
5	C	Male	<30 yrs			
6	C	Female	<30 yrs			

Ref #	Sea ice term	Definition
1	qarvitinniq	Ice that piles up, is attached to or clings to the shore. This ice is used to travel on by snowmobile when going along the coast. It only becomes dislodged by the biggest of the highest tides in June (typically second week of the month)[1]. *Alt. qingngusaliusijuq*[1]
2	qainnguq	Solid ice stuck to the edge of the shoreline[3]
3	aqiuppiniq	The line of ice that remains on rocks when the tide recedes[1]
4	illaujait	Icicles; crystals; splinter ice. This type of ice originates from upriver and becomes jumbled up with saltwater ice as it flows down rivers in spring. Very dangerous and unstable to walk on. Can be very long crystals. If one fell through this ice, one would need to use mitts to escape so as to avoid being cut[1,2,3]. *Alt. sikumi illaujait*[1]; *nilak* (Sch)
5	sikuutsuni tuvaq	Ice left on the flats by the tides; ice-laden shore[1]. *Alt. tinujaumajuk*[1]; *tinutsimajuk*[1]; *kutjutuq*[1]
6	akinnaq	Ice wall[1]
7	ivuniq	Ice piled up against shore[1]. *Alt. ilumunnatuq* (Sch)
8	qangngurusirtuq	An inlet that has frozen over. A long inlet is called *kangisukutaaq*[1]
9	napakkutait	Pointed ice[1]
10	tuvaq	Flat ocean, when frozen over. Sea ice. Flat ice unaffected by tides[1,2,3,4,5,6]. *Var. tuuvuq*[3]*; tuvaq* (Sch). *Alt. tariup sikunga*[1]; *sikuk tariuk*[2]; *tariuk sikunga*[3]
11	tuvaup sinaanga	Sea ice edge[1]
12	qunniq	Crack in ice with exposed water. When the tide rises, this crack disappears. This term can also be used to describe a big crack in a rock face on a mountain.[1,2,3] *Var. qunnik*[1]*; qumniq* (Sch). *Alt. qupisimajuk*[1]; *qupiniq killali*[2]
13	parniq	Lines or bands of ice with open water between them. Seafarers often have to travel some distance to make a course through each band[1]
14	qupinniq	Similar to *aajuraq (#35)*; but not a big crack. This crack opens and closes with the ebb and flow of the tides. A seal can also appear through this crack[1]. *Var. qupiniq*[1]
15	anarlugalak	Ice with mud on it; muddy ice[1]
16	qanittaq	Ice that forms overnight. Dangerous ice, thin ice. Also fresh-fallen snow on the ice. This term is mostly used along the floe edge environment[1]. *Alt. nutajak sattuk*[1]; *quuaak*[3]; *qinnuaq* (Sch)
17	qunniq	Edge of thin layer of ice (n.b., this term is also used to describe a crack)[2]. *Var. qinuit* (Sch)
18	sikuaq	Ice layer that forms over ice at high tide at floe edge[1,3]. Dangerous area; isn't ice, isn't water (Sch). *Alt. aqittuit*. Slush that comes up over sea ice[1]; *sikumi illaujait*[2]
19	sinaa	Floe edge; the last point at which water can reach[1,2,3,4]. *Var. sinaak*[4,5]; *sinaaq* (Sch). *Alt. tuvaup sinaanga*[1]; *imaup sinaanga*[1]
20	allutinni	Big hole in ice[3]. *Alt. kilaluk*[2]; *puturlak* (Sch)
21	manittuit	Rough ice that forms over the top of rocks or an island. Creates a mound or hill effect. This ice keeps piling up. It gets bigger as the tides become higher[1]. *Var. maniittuq/ manituk*[1]. *Alt. qullutuq manittuit*[1]; *maniittuit itjiniit*[2].
22	annisaq	Large floating ice, which is bigger than *apurtaniq*[1]
23	sikuup maniranga	Flat ice[2]
24	qaanik	Top layer of snow on ice[1]
25	tuvaauq	Sea ice, broken off [2]. *Alt. tuvaq nakattuk*[1]; *tuvaiqpuq* (Sch)
26	aannisaluk	Iceberg, general[2]. *Alt. qartuk*[3]; *piqalujaq* (Sch)
27	ilumaniq	An ice cave; an opening where one can go under the ice to collect clams from the sea-floor[1]
28	maniisiquit	Big pieces of rough ice[1]. *Var. manittualuit*[2].

Continued

Table 2
Continued

Ref #	Sea ice term	Definition
29	itjunik	A term used to describe the thickness of ice[1,3]. *Alt. sikuup itjuninga*[2]
30	titjauit	Ice crack that runs outwards from land into ice[1,2]. *Alt. aajuraq*[5]; *agiuppiniq* (Sch)
31	ittiniq	Rough ice beyond the shore (seaward) that continues to build up from tidal movements. Interface between land and sea ice.[1,2,3,4,5,6] *ittiniq* (Sch). *Var. ittiniit*[2,3,4,6]; *ittinik*[5]; *ittinnik*[5]; *ittinniq*[3]. *Alt. qaanuuk*[3].
32	kagpatuq	Chunks of ice, within open water, that rest against hard ice as they move back and forth with the tides[1]
33	uttuq	Animal/seal basking/resting on ice, rock or shore[1,3]. *Var. uuttuq* (Sch)
34	apurtaniq	Clumps of ice that settle in areas; these clumps move with the tides. Small, separate pieces of ice. Small chunks of ice; floating ice that constantly moves in and out of the bays in accordance with the tides[1]. *Alt. kinuatat*[1]; *puttaat*[1]; *siqaliat*[1]
35	aajuraq	Ice crack. Opens when the tide recedes and closes when the tide rises. A seal can appear through this hole in springtime and come to the surface [1,2,3,4,5,6]. *Var. aajurak*[4,5]; *aajuraq*[3,4,5]; *aajuraq* (Sch). *Alt. qunniq*; *qunnik*[1]; *qunniq*[3]; *qupisimajuk*[5]; *siku qupisimajuk*[6]
36	tuvarmik	Water that spills over onto *tuvaq*, such as when the tide rises[1]
37	qinnitarsiaq	The black layer of ice; the most bottom layer, as seen through a crack or when a hole is made through the ice[1]
38	aputijuk	Ice that collides with ice along sea edge[1]. *Alt. ullvanniq*[3]; *ivujut* (Sch)
39	igatutijuk	Ice that passes side-by-side [1]
40	masaligalaaq	Slushy ice[1]. *Alt. illujak*[1]; *maujaraq*[3]; *imarsuk*[4]; *matsaq* (Sch)
41	akluk	A breathing hole made by seals or narwhals. The seals keep this hole open during the winter[1]. *Var. alluk*; *aalluk*[3]; *Alt. puijiup anirtirivinga*[1]; *nulataq*[6]: *kiggulik* (Sch)
42	anirtirivik	Seal breathing hole in ice, snout can only pass through[3]. Alt. *qingainnavik*[6]
43	sikukallaq	Blocks of ice[2]. *Var. sikugippuq* (Sch)
44	sikuu ataa	Term to describe the underneath portion/space of ice[1]. *Alt. itjuniq*[1]; *tuvaup ataanga*[15]; *sikuup ataanga*[4]
45	kivakitak	Ice that takes its name from any seal of the seal or walrus family that sits on top of the floating ice. This term is not used when polar bears are on the ice[1]. *Var. kivakitaq*[1]
46	ivujuk	Thin ice layer that forms against blocks of floating ice[6]. Also ice that cracks in springtime. *Var. ivujuq*[1]; *ivujut (Sch). Alt. qanittaq*[1]; *tuvairpuq*[2]; *sikuliaq* (Sch); *qinnuaq* (Sch)
47	qingngugutik	Term used to describe the middle layer/level of sea ice (general)[1]. *Alt. sikuliak*[3]
48	tuvaup qaanga	Top layer/level of sea ice (general)[15]. *Var. tuvaup qanga*[1]
49*	ingirranitumariq	The highest tide of the year that will only dislodge the *qarvitinniq* ice from the shore (a typical high tide is called *ingirranittuq*)[1]
50*	ipiutaq	New ice[1]. *Alt. kakiattusia*[2]; *siku sattuk*[2]; *sikuak*[3]; *sikuliaq* (Sch)
51*	kakiatuq	Very white ice [1]
52*	kilaasuut	A period within the sea ice cycle (typically in spring) when holes begin to form within the land-fast sea ice region. Open water that appears when it is not cold [1]
53*	kilak	Hole in ice; often to describe a fishing hole[1,3,5]. *Var. kilaluk*[5]; *killaluk*[1]; *killaq*[5]. *Alt. nangianatuk*[1]; *alluak*[5,6]. See also: *akluk*
54*	kilak sikusuituk	Water that never freezes[1]. *Alt. sikuittuq*[2]; *aumanik*[5]; *aukkaniq* (Sch)
55*	mallu sikullu katinninga	Ice that conflates with water [1]
56*	nunguniq	Open water area that does not freeze because the current is too strong. There is an example of this at Abloviak Fiord[1]
57*	qarluq	The name for seaweed that is within the ice and becomes a part of ice[1]
58*	qarni sikumi	Old ice[2]. *Alt. atsuilik*[1]
59*	qitukkiqjuq	Thin ice that moves with the water, up and down, but does not move horizontally. This ice will not open[1]
60*	qunnialuk	Ice crack in middle of sea ice[1]
61*	sikuit	Ice that moves with the tides, which goes back and forth[1] See also: *utiqtaniq*
62*	sikuk	Ice, general term [1,2,3,4,5,6]. *Var. siku*; pure ice, good to melt for drinking. (Sch) *Alt. nilaq*[1]
63*	sinaanga	Edge of ice[2,3]
64*	siniritsijuk	Ice edge which envelopes open water [1]
65*	supiniq	The first break up of ice. This can only happen to a big river. The ice moves downstream at a rapid pace and generates much noise. This can be seen and heard on the George River and at Kuujjuaq in spring[1]
66*	tariungutsuni sikusimajuk	Seawater that transforms into ice [2]. *Alt. tuvaqpaaliuk*[2]
67*	tuvalirtuk	To travel from land to sea ice[1] *Alt. pijippuq* (Sch)
68*	tuvaq qupittuq	Sea ice cracked[1]

Continued

Table 2
Continued

Ref #	Sea ice term	Definition
69*	tuvvaviniit	An enormous piece of ice that has disappeared into the deep ocean. This ice will not return until the beginning of winter. A large piece of *tuvaq* that breaks away[1]
70*	utiqtaniq	Ice floating back and forth on the sea[1,2]. *Alt. aulaniq*[2]
71*	utirtaniit	Ice that comes and goes onto the shore and becomes thicker with every tidal movement[2]. *Var. utirtatut*[1]. *Alt. tulaktatuq*[1]

Table 3
Numbers and percentages of the 71 sea-ice terms identified, according to age and gender, by the study participants

Generation and gender		Number of respondents	Total of 71 possible terms identified	Percentage of 71 terms identified
A	Male	6	66	93.00
A	Female	3	24	33.80
B	Male	6	20	28.10
B	Female	6	7	9.85
C	Male	9	9	12.60
C	Female	4	8	11.20

relating to Inuit spatial cognition of the environment[9]. To 'standardize' the orthography provided by the participants against the orthography in a dictionary published over 25 years ago (the Inuktitut-English version) would ignore the changing nature of the language, and the growing use of different, yet related, terms to describe the same spatial features.

Types of sea ice

The number of terms identified in the study is significant, because it demonstrates that the Kangiqsualujjuamiut possess rich and varied knowledge about the sea ice environment. The terms were defined according to their associations with: seasonal and meteorological events (e.g., extreme cold); tidal and current patterns (e.g., spring tides); animal activities (e.g., seals basking); marine plant life (e.g., seaweed), and sea ice interfaces with other surfaces (e.g., mud, snow). Some terms had spatial connotations (e.g., offshore, edge, beneath) or were defined according to their colors (e.g., soiled, opaque), shapes (e.g., hole, crack), or formation (e.g., ice crystals, blocks, chunks). Furthermore, some ice forms were defined according to a sense of movement (e.g., up/down/across), or their ability to support weight (e.g., thickness). Although not explicit within the definitions, sea ice terms that referred to dangerous areas likely facilitate navigation and way-finding (as observed when accompanying hunters on excursions). Elsewhere in the North, Aporta (2002) and Laidler and Elee (2008) reported that travelling Inuit facilitate wayfinding by observing the positions of ice features in relation to the shore and to the floe edge.

If the 71 terms for sea ice that the cohort identified are viewed as a set, one may draw the conclusion that the Kangiqsualujjuamiut know their sea ice environment well. Their terminology, at least, remains a healthy subset within any broader collection of coastal knowledge. Since their sea ice environment is one of the most dynamic in the Canadian Arctic, it is perhaps not unexpected that the Kangiqsualujjuamiut possess a rich vocabulary for sea ice, just as the Australian Aborigines of the Central Desert region have various words to describe dust, or the Torres Strait Islanders and Papua New Guineans have multiple ways to describe reefs and cays (Nietschmann 1989; Mulrennan and Scott 2000; Sharp 2002; Levinson 2008). The connectedness of sea ice terms with other maritime features and phenomena suggests that knowledge of sea ice nomenclature is crucial to maintaining a holistic view, and appreciation, of the sea ice environment. The existence of highly descriptive terms such as *kivakitak* (a seal resting on floating ice) and *kilaasuut* (a period within the sea ice cycle when holes begin to form within the landfast sea ice region) is evidence that the Kangiqsualujjuamiut view sea ice as not just a generic

[9]The absence of a larger vocabulary of sea ice in the Schneider (1985) dictionary should not be regarded as a failing of that work. Schneider's dictionary was intended to be a broad reference, and not a lexicon of Inuit environmental terms. The terms in this article provide information about the sea ice environment beyond that found in Schneider.

substance, but as one that varies according to specific situations, settings, and other contexts.

Generational disparities in sea ice knowledge

This set of terms and definitions, if compared or combined with other sets of sea ice terms recorded elsewhere across the Arctic, might provide additional insight into the ways Inuit conceptualize and give meaning to their environment. Such an approach might enrich the understanding of Inuit knowledge of sea ice as a whole; but should consider disparities or inconsistencies in the nomenclature that exist between the generations of male and female Inuit groups studied, and between different Inuit groups across the Arctic. Any assessments of the robustness of their knowledge systems should include the efficacy of the *modes* of knowledge transmission, rather than solely cataloguing and comparing the knowledge that is imparted and retained (Haraway 1991). For example, if one were to generate a word list of Inuit sea ice terms based solely on the expert knowledge of Elders, it would not necessarily indicate the vitality, and the status, of sea ice knowledge within the broader community, in particular how the younger members of the community describe and communicate such information.

The shaded boxes at the bottom of Figure 6, and the related information in Table 2, reveal that only 4 of the 71 features identified were recognized by both genders of all three generations. Those features shown in Figure 6 include: *tuvaq* (flat, frozen ocean), *ittiniq* (rough ice), and *aajuraq* (crack). The fourth term, *sikuk* (a general term for ice), is listed in Table 2. These features are probably the most frequently identified sea ice terms because they are among the most conspicuous ice features around Kangiqsualujjuaq. However, to the travelling Inuk, knowledge of these four terms would impart only basic information. Missing from this most common vocabulary set are terms that would give a greater appreciation of the sea ice environment. These include terms relating to: dangerous ice; features beyond the floe edge; features that describe the layers of ice; holes and ice that is related to tidal, animal, or human activities. The English-equivalent terms that became apparent during the study will be used below to facilitate discussion. However, no attempt is made to impose a classification scheme on these terms as further research is required to determine whether a system of organizing and arranging sea ice phenomena is a feature of Kangiqsualujjuamiut knowledge base systems.

With only four sea ice terms in common use across all generations and genders of participants, it seems that the different groups possess different ways of knowing and seeing the sea ice environment. Knowledge about less-conspicuous forms of sea ice has greater importance for the older generations, who were not raised in settlements. The breakdown of terms identified by age and gender, shown in Table 3, was generated using information gathered from Table 2. Males from Generation A identified the most ice features, while females from Generation B identified the fewest. Knowledge of ice features by female hunters was much higher in Generation A, however, with only a third of the listed terms being identified. Pointing to what might be a gender-aligned system of sea ice knowledge, note that Table 3 indicates that the male participants in both Generation A and B identified more sea ice terms than did the female participants of the same generations. One reason for this difference may be that the women spend a significant amount of time looking after children and grandchildren in the home. Therefore, it is difficult for many women to find the time to travel and hunt, beyond occasional outings. According to many hunters, it is when out on the land and sea that knowledge about the environment is used and internalized for recall. This difference in knowledge and vocabulary also suggests that the female participants from Generation B do not envision the sea ice environment in the same way as do the male participants from Generation B. (This observation about envisioning the sea ice environment differently from the male participants of Generation A may also apply to *both* genders of Generation C, as each gender of that generation recognized fewer than 13 percent of the terms identified in the study).

The knowledge disparity between the males and females of Generation B is also likely a corollary of their hunting *patterns*. Female hunters today (as in the resettlement era) mainly interact with the sea ice environment close to shore, where they hunt geese and gather clams, fish, and seaweed. Seal hunting, however, continues to be a male-dominated activity. Because of this, male hunters are exposed to, and interact with, a greater sea ice environment than do female hunters. Consequently, it was not surprising to observe that fewer terms were identified by the female hunters of Generation B.

There was no disparity in knowledge between the males and females of Generation C. This may be because this generation has more equal access to school, and more equal participation in hunts, excursions, community activities, and programmes (such as Junior Rangers), than did the older generations.

The limited numbers of sea ice terms used by generations B and C suggest that the three generations of Kangiqsualujjuamiut each possess different interpretations of the sea ice environment. The findings of this study suggest that if a Kangiqsualujjuamiut Elder were to be accompanied by his son and grandson on a seal hunt to the *sinaa* (floe edge), each individual would interpret the journey differently. Upon departing the village by snowmobile, moving from land to sea, each hunter would be aware that he was crossing the threshold between these two realms. Each would encounter the ice formation known as *ittinniq*, a region of rough and undulating sea ice that forms a contiguous band around the coastline. The journey beyond this region onto *tuvaq*, the smooth sea ice, would also be known to each. But many of the specific sea ice features on *tuvaq* would not be equally recognized by each hunter. Examples include *napakkutait* (pointed ice) and *akinnaq* (ice wall), two features that might be used as navigation markers. Continuing seaward, the hunters might confront *aajuraq*, a generic term used to describe cracks in sea ice. Yet, as is apparent in Figure 6, only the Elder and his son would possess the vocabulary to determine whether *aajuraq* (#35) should be described as *qunniq* (#12), a crack or fissure with exposed water. If it was a crack that had not completely opened, only the Elder, according to this study, would possess the knowledge to describe it as *qupinniq* (#14). In another example, when at *sinaa*, only the Elder would have the capacity to describe the full gamut of sea ice features present in the open water beyond. The Elder might point to *apurtaniq*, which refers to clumps of ice. But his son and grandson would not know where to look, because their knowledge, as revealed in this study, is largely limited to sea ice features present along the shoreline. The Elder's knowledge of terms, however, extends to features across the entire sea ice environment.

If the task of transferring sea ice knowledge were left to the male experts from Generation B, it would appear from Table 3 that only one-third of all of the recorded sea ice terms could be passed on. Since the majority of the terms that Generation B identified were associated with the near-shore environment, individuals to whom they passed on their knowledge would not learn the terms and definitions associated with the *sinaa* environment.

This apparent diminution of sea ice knowledge might well be a consequence of fewer Inuit engaging with the *sinaa* environment. Using Inuit land-use and occupancy studies that recorded Kangiqsualujjuamiut patterns of travel on the ice over 25 years ago (Makivik Corporation 1985) for comparison, it is clear that current generations of Inuit do not venture onto the sea ice as frequently or range as widely as previous generations. Interacting less frequently with this zone, particularly the floe edge, means that hunters do not consistently use their knowledge of this space. This includes not only sea ice nomenclature, but also associated knowledge, such as animal behaviour and physiology, meteorology, navigation, and seasonal patterns. In this sense, the existing knowledge of sea ice nomenclature among the Kangiqsualujjuamiut may well be a barometer of the vitality of other forms of Inuit hunting and environmental knowledge. The skills required to hunt and travel in the sea ice environment depend upon a sound understanding of all these knowledge sets. On the other hand, hunting and travelling on the ice are necessary to keep the knowledge active and relevant.

The change in hunting behaviour over the past quarter of a century, from hunting across far reaches of sea ice to now carrying out most hunts closer to shore, is perhaps not the only factor affecting the transmission of sea ice knowledge. Maybe young and many middle-aged Kangiqsualujjuamiut hunters simply have not yet acquired the vocabulary that the Elders use to describe many of the features of the environment. Perhaps they never will. Perhaps they are familiar with such features, but did not indicate them in the study. At any rate, a loss of descriptive words probably means that future generations of Inuit will not have the capacity to understand the coast to the same degree as did their forebears.

In essence, any loss in knowledge of sea ice terms also means a loss in knowledge about the features associated with them. For example, the legacy of the sea ice as a provider for subsistence and navigational purposes would start to be lost. And, as sea ice terms associated with traditional stories diminish, so too could the stories themselves. This would lead to a loss in appreciation of the mythical aspects of the sea ice environment. For example, future generations may not perceive cracks in the sea ice as products of *Nanukulluk* stamping his feet. Or, when they happen upon holes in the ice, will they look around to ensure that *Mitilik* are not waiting nearby?

From a practical standpoint, a decline in sea ice terminology could well prove hazardous to travelling Inuit. This study suggests that young Kangiqsualujjuamiut are not familiar with the terms necessary to describe dangerous ice features such as *qanittaq* (ice that forms overnight), and *sikuaq* (ice that forms near a floe edge). Without the knowledge of such terms, and the settings in which they form, young hunters will be at risk of falling into the water. For survival alone, it is thus critical that the Kangiqsualujjuamiut know the full suite of Inuktitut sea ice terms.

It is important to recognize that some words that provide purposeful information are lost, and others are gained, in societies affected by acculturation to another society, such as the Inuit have been over at least the last 50 years. Their society has changed as a consequence of exposure to English as well as French religious, social, education, and technological systems. In light of Bruner's (1956, 191–199) observations that acculturated societies will 'retain the old [knowledge] and accept the new depending on which has greater usefulness', one might suppose that the Kangiqsualujjuamiut have retained only those words that are essential to carrying out their everyday lives. The Kangiqsualujjuamiut who have lived through the process of change from a preindustrial to a postindustrial society have faced many pressures that previous generations did not face. This period brought the challenges of living together in a settled community, and new occupations other than hunting due to the acceptance of a cash economy. This shift from living on the land to life in a village means that time spent on the land and sea is restricted to evenings, weekends, and holidays. This has undoubtedly affected the ability of some Kangiqsualujjuamiut to maintain their knowledge of the sea ice environment, since one's knowledge of the environment appears inextricably linked to the time spent, and the enduring associations forged with sea ice experts, in the field.

Such a decline in knowledge is not limited just to the Inuit. Many sea ice terms that were used by early English seafarers have disappeared from maritime lexicons. A similar decline in knowledge of maritime terms amongst American youth in New England was reported by Stilgoe (1994, 11). He stated that this loss of knowledge is leading to 'the death of topographical curiosity among the young', and has brought with it a transformed sense of connection to the coast. He argued that the loss of terms to describe features of coastal environments affects the way that coastlines are imagined, maintained, and respected. He further argued that the few terms that remained in use would embody renewed meanings, and a renewed sense of purpose, in the more modern context. Perhaps the same conclusion can be drawn if the Kangiqsualujjuamiut are to maintain their suite of sea ice terms.

To put Bruner's (1956) observations in context, sea ice terms that lack direct and immediate applicability will be the first to disappear from the Kangiqsualujjuamiut vocabulary. For the current set of terms to persist, hunters must maintain purposeful and spiritual attachments to the ice by frequently interacting with this environment. Subtle and obscure sea ice features, in particular, must be viewed (and understood) *in situ* in order to take on meaning, and to be spatially and seasonally conceptualized.

Getting the words right

As is apparent from Tables 2 and 3, this study revealed a variable orthography, and varying definitions of sea ice terms, within the cohort. Out of the set of 71 sea ice terms, 33 forms of sea ice had more than one name, and 14 terms had variable spellings. Terms with more than three alternative definitions (perhaps representing the most conspicuous ice features around Kangiqsualujjuaq) included: *sikuutsuni tuvaq* (ice on tidal flats); *sinaa* (floe edge); *apurtaniq* (ice clumps); *aajuraq* (crack); *masaligalaaq* (slushy ice); *sikuu ataa* (underneath portion of ice); and *ipiutaq* (new ice).

Some of these variations in spelling and definition, as discussed earlier, might be attributed to variations in literacy skills among the participants. Furthermore, one suspects that a hunter would have few occasions in which he or she is required to put sea ice terms on paper. Most would communicate such information orally, while on the sea. The irony is that the written word may be the only way of passing on this sea ice knowledge in the future, if the terms used are no longer transmitted orally.

Variations in spelling and definition might also be attributed to hunters applying the differing vocabularies used in the settings in which they were raised. For example, some of the Elders lived in hunting camps along the Labrador Coast in their youth. Inuktitut descriptions and terms for some sea ice features in Labrador are slightly different from the terms used on the Quebec side of the Quebec-Labrador Peninsula.

Another factor that may account for variations in the spelling of Inuktitut terms is the growing use of English and French in the community, particularly among Generations B and C. Furthermore, Inuktitut is not regularly used in online social networking sites, or in GPS devices, because such forums and media do not easily accommodate Inuktitut. This may explain why many young Kangiqsualujjuamiut participants identified sea ice terms using the English-equivalents first. This shift from Inuktitut to English or French in thinking, internalizing information, and communicating means that the Inuktitut vocabulary for the sea ice environment will ultimately be reduced, and that the nuances that these terms represent will also be lost.

If the 71 sea ice terms are to remain within the vocabulary of future generations of Kangiqsualujjuamiut, it is essential that a programme[10] be established to determine conventions for the accepted spellings and pronunciations of the terms and definitions. With fewer hunters exercising their knowledge about sea ice, and with the eventual passing of community Elders, there will be even more reliance on textbooks and illustrated dictionaries to transmit such knowledge in the future. Accurate recording of the sea ice lexicon in written and visual formats is essential if future generations of Kangiqsualujjuamiut are to retain their sense of identity and place in the polar coastal environment.

Conclusion

The evidence presented in this article demonstrates that the Kangiqsualujjuamiut community possesses a rich vocabulary for describing the sea ice environment. Many of the terms used take on definitions according to the season, and help describe the tides, animals, plants, and other phenomena in this setting. Some terms derive from characteristics of the ice itself, such as its shape, movement, color, or even its absence from an area. In addition, some features are defined according to their positions within the water column, or whether they are closer to or further away from the shoreline, or the floe edge. Sea ice knowledge is bound up with other forms of coastal knowledge, such as mythology, meteorology, and animal behaviour.

If this web of learning is to survive, it is essential that the existing sea ice vocabulary be preserved, and passed on to younger generations. However, the full suite of 71 sea ice terms identified were not known to the same extent by all generations of Kangiqsualujjuamiut. Significant disparities in knowledge between the Elders and the two younger generations suggest that current methods of transmitting knowledge within the community are not as effective as they once were when the Inuit shared this knowledge, principally through interactive, land- and sea-based activities.

[10]The Avataq Cultural Institute, recognizing the dynamism of Inuktitut, annually meets with representatives from each of Nunavik's communities to discuss the status of the language in the Territory, and to develop new Inuktitut words for technologies and situations as they arise. A natural extension of this forum would be to generate outreach programmes in communities to discuss new as well as previously recorded terms, especially where multiple ways of spelling and describing the same feature exist.

The ability to transfer knowledge depends upon the mechanisms by which that knowledge is transmitted, the efficacy of these mechanisms, and the number and availability of experts to educate nonexperts who want to become more knowledgeable. The strength, and the continuation, of knowledge (such as sea ice knowledge) is also dependent upon the social and environmental contexts in which potential recipients of that knowledge are situated, as this can affect their willingness, and ability, to learn and retain that knowledge.

To facilitate the transmission of Kangiqsualujjuamiut sea ice knowledge to future generations, the terminology gathered during this study will be contributed to the knowledge-amelioration programmes already operating within the community. Figure 6 will be offered to Kangiqsualujjuamiut educators for incorporation within the environmental and cultural modules at the local school.

If Kangiqsualujjuamiut sea ice knowledge were measured on the basis of the terms and definitions identified by the Elders alone, it would appear to be alive and healthy. But if it were measured against how such terms are known as one moves down the generational continuum, the results suggest that the knowledge base is in decline. This has important implications for the resiliency of the Inuktitut language, especially over the last half century. This study suggests that the Kangiqsualujjuamiut may have reached a crucial juncture in maintaining their wealth of meanings and understandings of Inuktitut terms related to sea ice; the question remains as to whether this form of environmental knowledge will continue to be a feature of their language in the long term.

Other important questions are raised by the fact that the youth of Kangiqsualujjuamiut have relatively little knowledge about the sea ice environment. With only a handful of words to describe it, what are they to make of the sea ice environment? How will they and future generations connect to this setting, if those words—the words that give it richness, variety, and a sense of heritage and usefulness—have all but disappeared?

Although it would be a challenge, there is still the possibility (if desired by the community) that this information can be transmitted to the younger generations while the Inuit Elders are still alive. As the results presented here demonstrate, it will be important to use an intergenerational approach to establish a base for this knowledge, if it is to be fully understood. This could occur, for example, through hunts taking place on sea ice more often, or by carrying out other activities on the ice that involve the acknowledged experts on this substance. However, the Elders who could transmit this knowledge are reaching a phase in life where it is becoming more difficult to spend long hours out on the ice. Consequently, as the time spent with Elders on the ice becomes more limited, it will be necessary for new forums to emerge whereby their knowledge can be meaningfully and purposefully exchanged.

By closing these cracks that exist in the knowledge, the full suite of words and definitions still available may yet be passed on to younger generations, thus enabling them to recognize and appreciate the sea ice environment more fully, as their ancestors have known it.

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